Rithindatta Gundu

AI/ML Engineer

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Professional Summary

AI/ML Engineer with 3+ years of experience leveraging C, C++, C#, and Python to develop scalable, secure, high-availability ML models and distributed systems. Expertise in building microservices for AI workflows, integrating LLMs (OpenAI APIs) with NLP pipelines, and deploying on Google Cloud/AWS using Kubernetes and CI/CD. Strong in cloud security, applied cryptography, OS internals, and performance optimization, with proven results in reducing manual processing by 70% and deployment times by 50%.

Technical Skills

- **Languages**: C, C++, C#, Python, Java, JavaScript, SQL
- **AI/ML Frameworks**: TensorFlow, PyTorch, Hugging Face Transformers, Scikit-learn, OpenAI APIs; NLP, LLM Integration, Deep Learning (CNNs, RNNs)
- **Distributed & Cloud**: Google Cloud (Compute Engine, Cloud Run, Vertex AI), AWS (EC2, S3, Lambda, SageMaker), Kubernetes, Docker, GitHub Actions, Terraform
- **Security & Systems**: Windows OS Internals, Applied Cryptography (AES, RSA), VM Provisioning, Secure APIs (OAuth, JWT, gRPC)
- **Frameworks**: FastAPI, Node.js (Express), React, Spring Boot, .NET Core
- **Databases**: PostgreSQL, MySQL, MongoDB, Redis
- **Monitoring & Debugging**: Prometheus, Grafana, EFK Stack (Elasticsearch, Fluentd, Kibana), gdb, Valgrind, Wireshark
- **Practices**: Agile/Scrum, CI/CD (Jenkins, GitHub Actions), Test-Driven Development (TDD), Secure Coding, Model Deployment (MLOps)

Professional Experience

AI Software Engineer | Wells Fargo, San Francisco, CA (Remote/Hybrid) *September 2024 – Present*

- Developed secure backend ML services in C++ and Python for AI-driven automation in financial risk workflows, integrating OpenAI LLMs for NLP-based fraud detection and reducing manual processing by 70% across 500K+ transactions.
- Designed and tested RESTful/gRPC APIs for VM provisioning and encryption pipelines using Kubernetes and GCP Vertex AI, improving inference efficiency by 40% and scaling to 10K+ daily model requests with 99.9% uptime.
- Containerized AI microservices with Docker and deployed to hybrid Google Cloud/AWS environments via GitHub Actions CI/CD, reducing deployment time by 50% and enabling seamless A/B testing for model variants.
- Collaborated with cross-functional teams (ML, security, DevOps) to embed cloud security best practices, including RBAC, data encryption at rest/transit, and compliance audits (SOC 2, PCI-DSS), mitigating vulnerabilities in 25+ production services.
- Optimized model performance using Valgrind/gdb for memory profiling and Prometheus/Grafana for monitoring, cutting latency by 25% in real-time anomaly detection systems.

Software Engineer | Maxgen Technologies Pvt. Ltd., Hyderabad, India *September 2021 – September 2023* (2 years)

- Engineered scalable AI/ML platforms using C#, Python (PyTorch/TensorFlow), and PostgreSQL for enterprise automation, supporting 50K+ users with ML models for predictive analytics and achieving 95% accuracy in workflow optimization.
- Built secure GraphQL/REST APIs with RBAC/JWT authentication and LLM integration (Hugging Face), deploying on AWS SageMaker for NLP tasks like sentiment analysis, improving system scalability by 35% during peak loads.
- Led development of cloud-native microservices on AWS EKS/GCP, provisioning VMs with Terraform/Docker for ML training pipelines, reducing infrastructure costs by 30% through auto-scaling and resource optimization.
- Implemented real-time monitoring for AI models with Prometheus/Grafana and EFK Stack, reducing downtime by 25% and enabling proactive debugging for distributed inference across multi-region setups.
- Conducted code reviews and TDD with Pytest/JUnit, achieving 90%+ coverage; optimized C++ modules for OS-level ML acceleration, enhancing data processing by 40% in cryptography-integrated systems.
- Collaborated in Agile sprints via Jira/Confluence, delivering secure AI features for finance/healthcare clients, including confidential computing prototypes with TEEs.

Education

Master of Science in Computer Science Seattle University, Seattle, WA *September 2023 – June 2025 (Expected)*

- Roles: Graduate Assistant (GA) for Dean of Students, SU Assessment Committee Member, ACM Representative

Bachelor of Technology in Computer Science CMR College of Engineering & Technology, Hyderabad, India *August 2019 – May 2023*

Projects

Confidential Computing Pipeline for Secure ML Inference | Python, GCP Confidential VMs, TensorFlow/PyTorch, Hugging Face | GitHub: [Link]

- Architected secure ML pipeline using GCP Confidential VMs for encrypted training/inference on sensitive datasets (1M+ records), integrating OpenAI APIs for LLM-based NLP fraud detection with 92% accuracy.
- Implemented TEEs with Rust enclaves for zero-knowledge proofs and cryptography (AES-256), reducing latency by 25% while maintaining FIPS 140-2 compliance.
- Deployed with Kubernetes/Docker; monitored via Prometheus/Grafana for scalable, secure AI ops.

Distributed VM Provisioning System for AI Workloads | C++, Kubernetes, AWS Lambda/SageMaker, Terraform | GitHub: [Link]

- Built C++ orchestrator provisioning 100+ VMs for ML training on AWS/GCP, using Terraform IaC and Lambda triggers to scale 5K+ requests/hour with 98% success rate.
- Integrated secure APIs with JWT/OAuth and ECDSA key management, improving provisioning speed by 40% for distributed AI clusters.
- Used EFK Stack/Valgrind for logging/debugging, reducing errors by 30% in high-availability setups.

NLP-Powered Secure Chatbot for Compliance Auditing | Python, FastAPI, PostgreSQL, BERT/PyTorch | GitHub: [Link]

- Developed FastAPI backend with React frontend for compliance chatbot using BERT for NLP on encrypted Redis/PostgreSQL, handling 1K+ queries/day with 95% accuracy.
- Applied secure coding (sanitization, rate limiting) and GitHub Actions CI/CD, optimizing for low-latency in distributed environments.